Emna Mastouri

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EDUCATION

PhD in computer science 2026 – present

Faculty of Sciences of Tunis, LIMTIC laboratory

- Configuring adaptive learning systems under uncertainty, with validity and stability constraints.

Research Master's in computer science (software engineering) 2024 – 2026

Higher Institute of Computer Science (ISI), University of Tunis El Manar

- Thesis: a neuro-symbolic approach to personalised mathematics tutoring for the Tunisian baccalaureate.

Bachelor's degree in information and communication sciences and technologies 2021 – 2024

Faculty of Sciences of Tunis, University of Tunis El Manar

Baccalaureate in experimental sciences 2021

Lycée HHAW

EXPERIENCE

Part-time lecturer Ongoing

Higher Institute of Computer Science (ISI), University of Tunis El Manar

End-of-studies internship February – May 2024

Tunisie Télécom

- Developed a web application for monitoring computer networks.
- Technologies: Nagios XI, GNS3, VMware, Winbox, Angular, Node.js, MongoDB

Internship Summer 2022

Tunisie Télécom, Central Service Management Unit, Kasbah

- Maintenance of network equipment and transmission media.

ACADEMIC PROJECTS

Quality of LLM-generated vs. human-written code *Python, SonarQube*

A comparison of code generated by language models with human-written code, looking at correctness, complexity and security warnings.

Automated test generation and mutation testing *Pynguin, mutmut*

Evaluation of two test generation tools on open-source projects, using code coverage and mutation score.

Evolution of code smells in open-source projects *PyDriller, SonarQube*

Tracking code smells across several releases of open-source projects, and studying how they relate to bug-fix commits.

SKILLS

AI and language models	Python, LLM, RAG, multi-agent systems, SymPy, Streamlit
Research tools (PhD)	OR-Tools CP-SAT, MaxSAT, Knowledge tracing
Software quality and testing	SonarQube, PyDriller, Pynguin, mutmut
Web development	Angular, Node.js, MongoDB
Networking	Cisco (routers and switches), Nagios, GNS3, SD-WAN, IPv4 / IPv6, dynamic routing
Embedded systems	Arduino, ESP, STM, Raspberry Pi
Personal strengths	communication, interpersonal skills, adaptability, logical thinking, autonomy

LANGUAGES

Arabic: native | English: fluent | French: intermediate